WINTECK

OOPS Question – 23/05/2026

1. What is a class in Python? How do you define and create an object from a class?
2. Can you explain the difference between a class and an object in Python? Provide examples.
3. what is __init__ and __new__ in python
4. Can we create class without __init__ and __new__?
5. Can we use return in __init__method?
6. What are the 4 fundamental principles of OOPS
7. What is inheritance and what are different types of inheritance explain with examples
8. What is the use of super() keyword
9. What is MRO and how it be related to multiple inheritance
10. Implement two classes with same method name and inherit in child class , create an object for class3 and call methods from two parent classes
11. What are class variables & instance variables explain with examples
12. What is static method and class method why we need to use these two methods
13. What is Encapsulation explain with examples
14. What are private , public and protected attributes explain with examples
15. What is polymorphism explain with different examples
16. What is the difference between method overloading and overriding explain with exmaples
17. What is abstraction explain the uses of abstraction in python
18. What are getter and setter methods, and why are they important in encapsulation?
19. What is the purpose of the @abstractmethod decorator?
20. How does Python determine which method to call when a method is inherited from multiple parent classes?
21. What is multiple inheritance? Give a code example. Ex with Super and MRO
22. Can a class inherit from multiple classes?
23. How does Python's garbage collection affect objects?
24. Create decorator to calculate actual time taken by original function?
25. what is the difference between encapsulation and abstraction explain with example?
26. Can we decorate same function with two different decorators explain with example?
27. Create a class Student with attributes:
    I. Name
    II. Roll_number
    III. Marks

    Add a method to display student details?

28. Create a class BankAccount with:

I. Account holder name

II. Balance

Add methods:

Deposit

Withdraw

Display balance

Implement classes for these two?

29. Constructors and Instance Variables

Create a class Employee using a constructor to initialize:

Employee id

Name

Salary

Create a class Laptop that stores:

Brand

RAM

Processor

30. Encapsulation

Create a class ATM where balance is private.

Add methods to deposit and withdraw money safely.

Create a class Password Manager.

Store password as a private attribute.

Create methods to update and validate password.

Create a class Student.

Make marks private.

Add getter and setter methods.

31. Inheritance

Create a parent class Animal with method sound().

Create child classes Dog and Cat.

Override the sound() method.

Create a class Vehicle.

Add attributes: speed and fuel.

Create child class Bike.